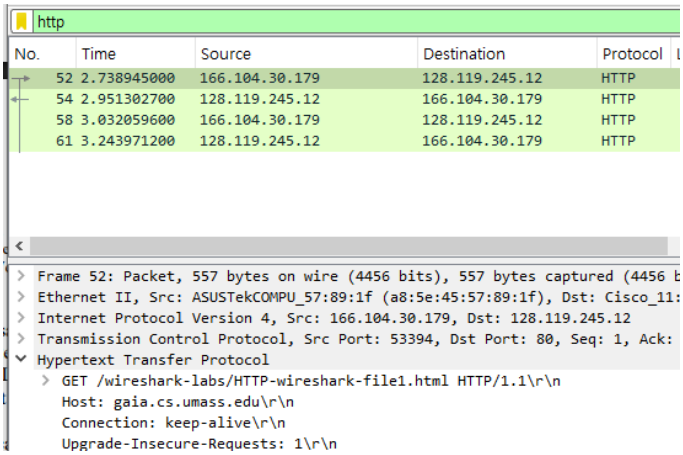
WireShark Lab #2 - HTTP

학번: 2022063672

이름: 박수현

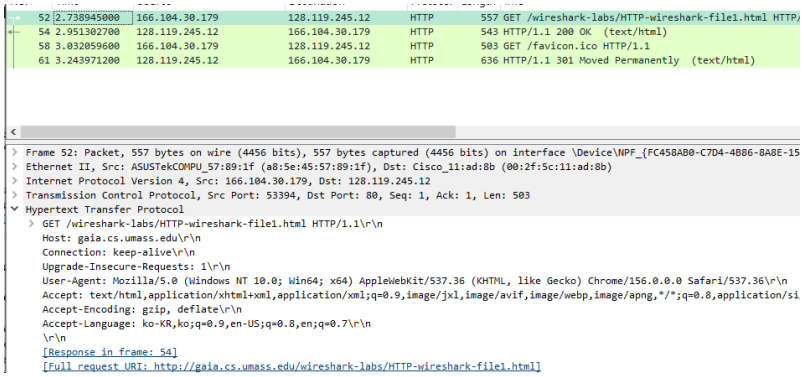
1. Is your browser running HTTP version 1.0, 1.1, or 2? What version of HTTP is the server running?

서버와 클라이언트 모두 http1.1을 사용중이다.



2. What languages (if any) does your browser indicate that it can accept to the server?

한국어와 영어 둘 다 accept 가능하다



3. What is the IP address of your computer? What is the IP address of the gaia.cs.umass.edu server?

클라이언트 IP: 166.104.30.179

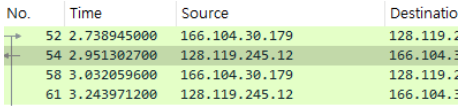
서버 IP: 128.119.245.12

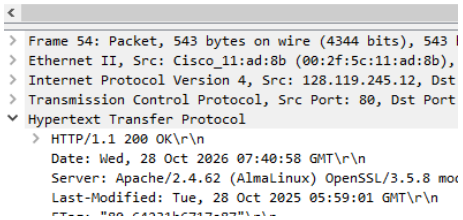
4. What is the status code returned from the server to your browser?

200 OK

5. When was the HTML file that you are retrieving last modified at the server?

Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT





6. How many bytes of content are being returned to your browser?

128 bytes

7. By inspecting the raw data in the packet content window, do you see any headers within the data that are not displayed in the packet-listing window? If so, name one.

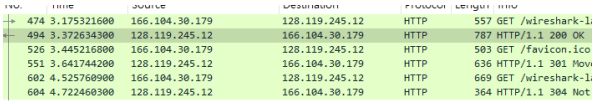
날짜, 서버 정보 등 다양한 종류의 헤더가 포함되어 있음



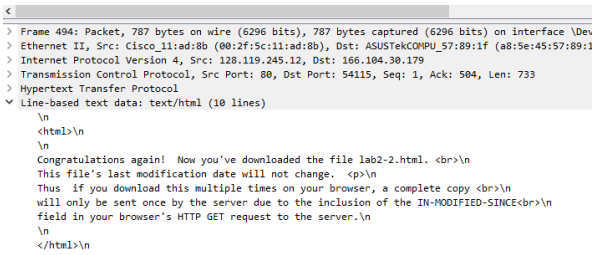
8. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?

브라우저 캐시를 삭제한 상태이기 때문에 IF-MODIFIED-SINCE 항목이 보이지 않는다.

9. Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?

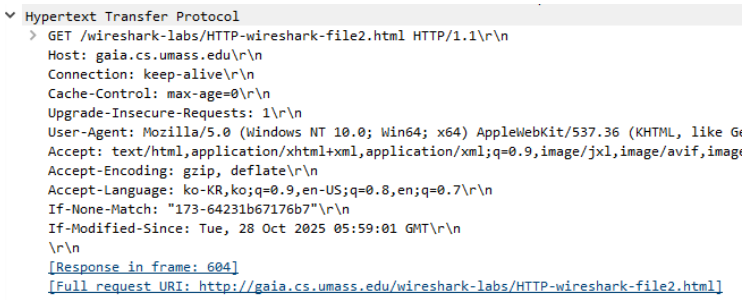


html 컨텐츠 파일이 포함되어 있는 것을 확인할 수 있다.



10. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?

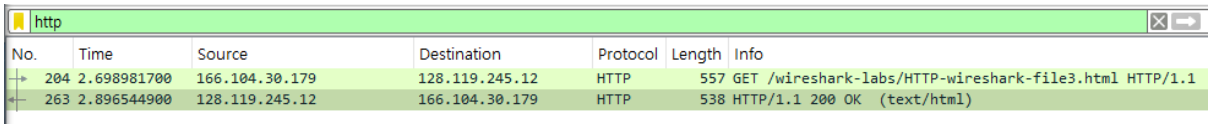
서버가 첫 번째 요청에서 전송한 파일의 수정 시각을 함께 전송하고, 두 번째 요청에서 그 시각을 IF-MODIFIED-SINCE 헤더에 담아 보낸다



11. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

304 Not Modified를 리턴한다. 캐시가 유효한 것이 확인되었기 때문에 그 응답 패킷에는 콘텐츠 파일이 포함되지 않는다.

12. How many HTTP GET request messages did your browser send? Which packet number in the trace contains the GET message for the Bill or Rights?



1개의 http GET 요청 송신. 204번 패킷이 GET 메시지를 송신했다.

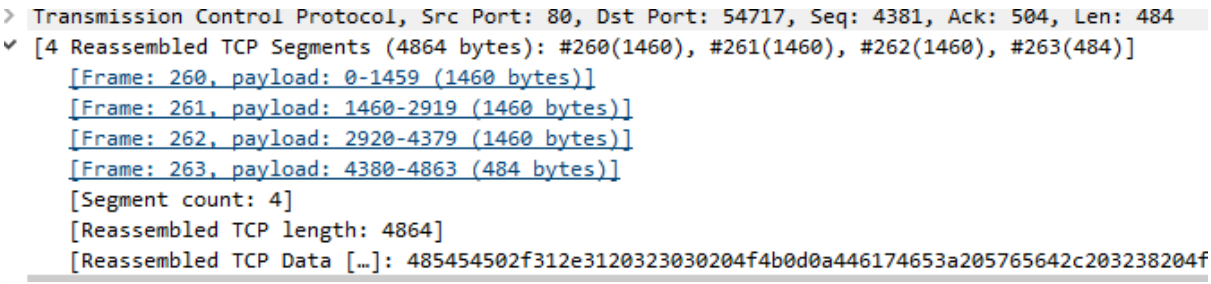
13. Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?

263번 패킷이 HTTP GET에 대한 응답 status code 200 OK를 담고 있다.

14. What is the status code and phrase in the response?

200 OK

15. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights?



4개의 TCP segments 로 나뉘어 수신되었다.

16. How many HTTP GET request messages did your browser send? To which Internet addresses were these GET requests sent?

2개의 GET 요청을 송신했다. 두 요청 모두 IP:128.119.245.12를 목적지로 송신되었다.

17. Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two web sites in parallel? Explain.

2번째 이미지에 대한 URL이 보안 연결을 지원하지 않아 학교 접근망이 2번째 이미지 수신을 차단하여서 실제 패킷 동작을 판단하기 어려웠다. 만약 정상적으로 접근이 가능했다면 요청을 병렬로 전송했을 것이다.

